DDI R-UI: a tool for describing data in R

Domingo Scisci

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Outline

1. **THE CONTEXT**
   a. UniData - Bicocca Data Archive
   b. UniData and R

2. **DDI R-UI**
   a. What it is
   b. Features

3. **DDI R-UI in the future**
   a. Short/medium term
   b. Long term

The Context - UniData

UniData - Bicocca Data Archive

Interdepartmental Centre, born in 2015, that aims to become the Italian Service Provider in CESSDA

- ADPSS-Sociodata legacy
- small (temporary) staff (2 people):
  - no division of roles
  - high workload

Need of tools that help us to

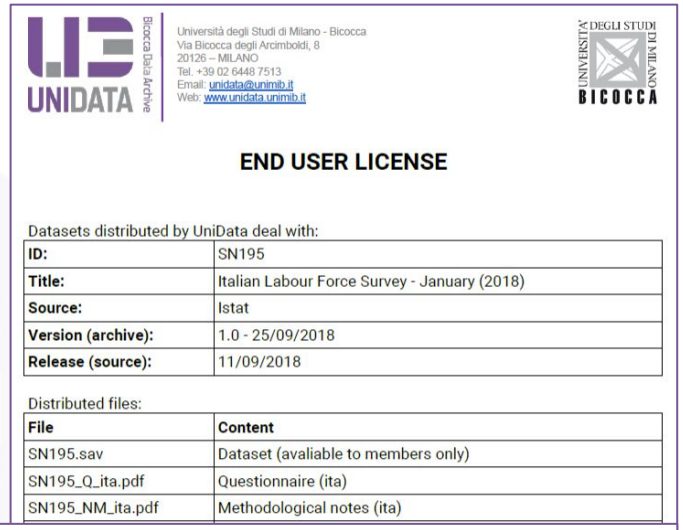
- **automate some procedures**
- **prevent errors**
- **make the work easier**

The Context - UniData and R

In 2015 we decided to use R for our work.

Since then, a R library has been developing to manage the ingest process:

- **data** management
- **labels** management
- calculate/verify **checksums**
- **codebook** (from data to PDF file)
- **license** (from DDI to PDF file)
- SPSS/Stata **syntaxes**
- **DDI file/dataDscr** from data file(s)



UNIDATA Bicocca Data Archive

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Tel. +39 02 6448 7513
Email: unidata@unimib.it
Web: www.unidata.unimib.it

UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

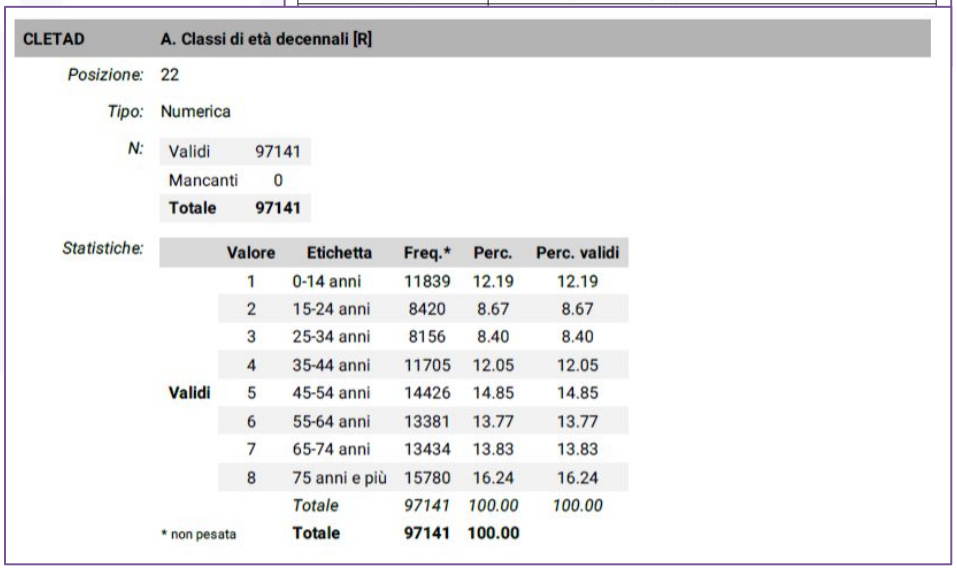
END USER LICENSE

Datasets distributed by UniData deal with:

ID:	SN195
Title:	Italian Labour Force Survey - January (2018)
Source:	Istat
Version (archive):	1.0 - 25/09/2018
Release (source):	11/09/2018

Distributed files:

File	Content
SN195.sav	Dataset (avaliable to members only)
SN195_Q_ita.pdf	Questionnaire (ita)
SN195_NM_ita.pdf	Methodological notes (ita)



CLETAD **A. Classi di età decennali [R]**

Posizione: 22

Tipo: Numerica

N:

Validi	97141
Mancanti	0
Totale	**97141**

Statistiche:

	Valore	Etichetta	Freq.*	Perc.	Perc. validi
	1	0-14 anni	11839	12.19	12.19
	2	15-24 anni	8420	8.67	8.67
	3	25-34 anni	8156	8.40	8.40
	4	35-44 anni	11705	12.05	12.05
Validi	5	45-54 anni	14426	14.85	14.85
	6	55-64 anni	13381	13.77	13.77
	7	65-74 anni	13434	13.83	13.83
	8	75 anni e più	15780	16.24	16.24
		Totale	*97141*	*100.00*	*100.00*
	* non pesata	**Totale**	**97141**	**100.00**	

DDI R-UI

- so far docDscr and StdyDscr sections are filled in Wordpress, then merged with fileDscr and dataDscr ones in R
- next year we will move from Wordpress to Dataverse (DataverseEU project) platform

A new tool to

- **manage all metadata (doc/stdy/file/dataDscr) in the same environment**
- **have a clear separation between ingest (R) and distribution (now: WP, future: DV) processes/platforms**

What DDI R-UI is

User Interface for DDI-C 2.5

- web **form**
- developed with **R** (**Shiny**) → little knowledge of JS/HTML/CSS required
- **no** need any other **software** (web server, db)
- it can be used **inside RStudio** locally or in a web browser from a **remote** server
- **customizable** → fields from a config file
- **multilanguage**
- **multiplatform**

This is the title of the study (2018)

SN000

1.0

DDI-UI | Summary | docDscr | stdyDscr | Export

- > Study Title
- > Study Subtitle
- > Study Translated Title
- *Study Identifiers*
  - > IDNo
  - > IDNo - Agency
- **Study Authors Section**
  - *Study Authors*
    - > Author
    - > Author - Affiliation
    - > Author
    - > Author - Affiliation
  - *Author - Identifier*
    - > Value
    - > Scheme
- *Contacts*
  - > Contact Name
  - > Contact Affiliation

Study Title

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Study Subtitle

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Study Translated Title

Study Authors Section

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Study Authors

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Author

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Author - Affiliation

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Author

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Author - Affiliation

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Author - Identifier

Value

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Scheme

ORCID

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Topic Classification

ECONO

- ECONOMIA
- ECONOMIA - economia rurale
- ECONOMIA - politica economica
- ECONOMIA - sistemi economici e sviluppo
- ECONOMIA - consumi/comportamenti di consumo
- ECONOMIA - condizione economica e indicatori
- ECONOMIA - reddito, proprietà e investimenti

DDI Title

Documentazione XML DDI-C 2.5 per l'indagine

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed nisl augue, euismod id f
Curabitur tincidunt dolor blandit, sagittis nisl ut, finibus ipsum. Morbi semper dictum
porta risus fermentum euismod.

DDI Translated Title

XML DDI-C 2.5 Documentation for study

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DDI IDNo

DDI-

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DDI IDNo - Agency

UniData

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Sed nisl augue, euismod id facilisis et, rutrum sed nisi.

This is the title of the study (2018)

SN000

1.0

DDI-UI | Summary | docDscr | stdyDscr | Export

Here you can download metadata in XML and JSON formats

Download XML

Download JSON

DDI R-UI - Under the hood

Config File:
- type (section, group, single)
- Attributes (repeatable, multilingual, mandatory)
- CSS
- XML (XPath)
- JSON

Imported as a **list** of R6 objects (3 classes: section, group, single - hierarchical).

Create the UI (single/multiple fields)

for each object:

new object for each field (stored in a list):
- value
- observer
- XML
- JSON

create a RDS file (resume)

export to XML

export to JSON

DDI R-UI in the future

Short/medium term

- better name???
- testing and release of the first version soon
- features available in 1.0: compile, export, and validate the DDI; JSON export
- collaboration with other projects (DDIwR)
- sources on public repository?
- export DDI in PDF format

DDI R-UI in the future

Long term

- re-writing of the front end outside R (JavaScript/Python) → separate back end from front end
- store information in an external DB
- create a stand-alone app (nodeJS)?

THANK YOU!

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1.0 - 2018-12-04